

F1 Data Lab - V1 Production Document

1. Vision

F1 Data Lab V1 is an opinionated, data-driven analytics platform launching at the start of the 2026 Formula 1 season. The goal is to introduce bold, debate-generating performance metrics backed by transparent methodology. The product positions itself as an analytical challenger brand within F1 discourse.

2. Core Product Scope (V1)

V1 is strictly limited in scope to ensure on-time launch by March 8. The application includes: - 3 custom performance metrics - Driver leaderboard - Driver deep dive page - Methodology transparency page - Static historical dataset (2023–2025 seasons)

3. Custom Metrics Overview

Metric 1: Overperformance Score Measures driver grid-to-finish delta adjusted by teammate baseline. Metric 2: Chaos Score Measures performance in races with safety cars, high DNFs, or wet conditions. Metric 3: Sunday Score Measures race-day execution including position gain and consistency.

4. Data Architecture

Data Sources: - Ergast API for historical results - Manual preprocessing for DNF categorization - Pandas for transformation and aggregation Processing Pipeline: - Extract race data - Normalize grid and finishing positions - Compute per-race metric components - Aggregate seasonal averages - Cache processed dataset for frontend use

5. Frontend Requirements

Design Direction: Dark, aggressive, esports-inspired UI. Pages: - Home / Leaderboard - Driver Profile Page - Metric Explanation Page UI Components: - Animated sortable leaderboard - Radar chart for driver metrics - Trend line graph per season - Shareable stat cards

6. Development Timeline (6 Days)

Day 1-2: Data ingestion and preprocessing Day 3: Metric modeling and testing Day 4-5: Frontend UI build and visualization Day 6: Polish, deployment, and social preview assets

7. Deployment & Tech Stack

Backend: Python, Pandas Frontend: Streamlit or Next.js Hosting: Vercel (Next.js) or Streamlit Cloud Version Control: GitHub

8. Launch Strategy

Launch timing: March 8, 2026 (Season Opener) Initial content strategy: - Pre-season ranking post - "Most Overrated Driver Entering 2026" - "Chaos King Ranking" - Weekly update after each Grand Prix

9. Risk Assessment

Risks: - Metric controversy - Data integrity issues - Low initial traffic Mitigation: - Full methodology transparency - Clear metric explanations - Community engagement and iteration

10. Success Criteria

Short Term: - 500–1000 visits during launch week - Community discussion engagement Medium Term: - Weekly recurring traffic - Recognition in F1 online communities